	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering (PCCoE) (An Autonomous Institute) Affiliated to Savitribai Phule Pune University(SPPU)	

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Lab 6: For given case study prepare the UML diagrams: Class Diagram , Activity Diagram, Sequence Diagram, Package and deployment diagram

1. Aim

To study, understand, and represent different aspects of a software system using UML (Unified Modeling Language) diagrams, including structural and behavioral views such as Class, Activity, Sequence, Package, and Deployment diagrams.

2. Problem Statement

The tourism industry involves multiple services such as tour package management, customer bookings, payments, and communication between users and administrators. In traditional systems, these processes are often handled manually or through disconnected platforms, leading to inefficiencies such as data inconsistency, lack of real-time updates, poor user experience, and difficulty in managing large volumes of information.

There is a need for an integrated and automated system that can efficiently manage all tourism-related activities in a centralized manner. The system should allow tourists to browse available tour packages, view details, make bookings, and perform secure online payments. At the same time, administrators should be able to create, update, and manage tour packages, monitor bookings, and handle customer interactions effectively.

To address these challenges, the TourMate Tourism Management System is proposed. This system aims to provide a user-friendly platform that streamlines the entire tourism management process, improves communication between stakeholders, ensures data accuracy, and enhances overall operational efficiency.

Theory

1. Class Diagram

A **Class Diagram** is a structural diagram that represents the static structure of a system. It shows:

- Classes
- Attributes (variables)
- Methods (functions)
- Relationships between classes

Key Components:

- **Class**: Represented by a rectangle divided into three parts
- **Attributes**: Data members of a class
- **Methods**: Functions of a class
- **Relationships**:
    - Association
    - Inheritance (Generalization)
    - Aggregation
    - Composition

Purpose:

It is used to design the system architecture and define how different classes interact.

2. Activity Diagram

An **Activity Diagram** represents the flow of activities or actions in a system. It is similar to a flowchart.

Key Components:

- **Start Node (●)**
- **Activity/Action**
- **Decision Node (◇)**
- **Control Flow Arrows**
- **End Node**

Purpose:

- To model workflows
- To represent business processes
- To show conditional logic and parallel processes

3. Sequence Diagram

A **Sequence Diagram** shows how objects interact with each other in a specific sequence of time.

Key Components:

- **Actors/Objects**
- **Lifelines** (vertical dashed lines)
- **Messages** (arrows)
- **Activation Bars**

Purpose:

- To show the order of operations
- To represent communication between objects
- To model real-time interactions

4. Package Diagram

A **Package Diagram** is used to organize large systems into smaller, manageable groups called packages.

Key Components:

- **Packages** (folders)
- **Dependencies** (arrows between packages)

Purpose:

- To manage complexity
- To group related classes
- To show dependencies between modules

5. Deployment Diagram

A **Deployment Diagram** represents the physical deployment of software on hardware.

Key Components:

- **Nodes** (servers, devices)
- **Artifacts** (software components)
- **Communication paths**

Purpose:

- To show system architecture
- To represent hardware-software mapping
- To visualize system deployment

Conclusion

These UML diagrams collectively help in understanding both the **structure and behavior** of a system. They are essential tools for software developers to design, analyze, and communicate system architecture effectively.